AQUABOTS Hull Wiring Integration Guide

Team 26061 — ENGR 498B Spring 2026
Rev WG-1 | 2026-03-17 | Owner: John
Derived from Changelog Rev ISR-4b + subsequent conversation updates

IMPORTANT — Document Accuracy Note: The changelog Section 1.3 branch table still shows 16/18 AWG for some branches. The ISR slides show MRBF 150A and MIDI 125A. Both are **stale**. This guide reflects actual current specs from all revisions through ISR-4b plus the 12 AWG standardization decision. When in doubt, **this document governs**.

1 — Hull Layout Reference

Hull: teardrop ~1m × 0.4m. Pointy end = bow, wide end = stern. 12×24" aluminum plate in forward-mid section (velcro mounted). ESP32-S3 on Meshnology breakout board on hull cap top plate (preserves IPEX antenna cable).

Zone	Components	Position
Top Deck (bow-mid)	Starlink Mini, GPS, Camera	Above plate, sky-facing
Forward on Plate	(+) Busbar, (–) Busbar, MRBF, Kill SW, MIDI, ATO	First 6" of plate
Mid Plate	12V Buck, 5V Buck, 5V Pwr Mod, BEC, ESC	Middle 8" of plate
Aft on Plate	Jetson Orin Nano, Pixhawk 4X, ESP32-S3	Last 6" of plate
External Bow Mount	pH, Temp, TDS, Ultrasonic (submerged)	Below waterline, bow
Aft / Below Plate	Battery (demo or production)	Widest hull section
Stern	Motor (5080), Rudder Servo	Stern centerline

2 — Power Wiring

2.1 — Main Trunk (4 AWG)

ID	From	To	Gauge	Length	Color	Fuse	Ring Terminal
P1	Batt (+)	MRBF Holder In	4 AWG	6"	RED	MRBF 100A/40A	SC16-8 (M8)
P2	MRBF Out	Kill Switch	4 AWG	6"	RED	—	SC16-6 (1/4")
P3	Kill Switch	(+) Busbar	4 AWG	6"	RED	—	SC16-6 (1/4")
P4	Batt (–)	(–) Busbar	4 AWG	12"	BLK	—	SC16-6 (1/4")

Ring terminal note: 4 AWG wire is actually 19mm² welding cable (undersized vs 21.15mm²). Use SC16-8 for M8 battery studs and SC16-6 for 1/4" busbar studs. Do NOT use SC25.

Demo adapter pigtail (removable):

ID	From	To	Gauge	Length	Color	Connector
D1	XT90■ (+)	MRBF input stud	10–12 AWG	6–8"	RED	XT90■→ring
D2	XT90■ (–)	(–) Busbar	10–12 AWG	6–8"	BLK	XT90■→ring

2.2 — Motor Branch (8 AWG)

ID	From	To	Gauge	Length	Fuse	Notes
P5	(+) Busbar	ESC Input (+)	8 AWG	6"	MIDI 60A	RED. Through MIDI block.
P6	(–) Busbar	ESC Input (–)	8 AWG	6"	—	BLK. To ATO (–) bus.

ID	From	To	Gauge	Length	Fuse	Notes
P17	ESC (3ϕ)	Motor 5080	8 AWG	12–18"	—	3 wires. Bullet 4mm. Centerline.

Routing: ESC-to-motor 8 AWG along hull centerline in dedicated split loom, 12–18"+ from ESP32 and Pixhawk. This is the primary EMI source.

2.3 — Busbar Branch Circuits (ALL 12 AWG)

All non-motor busbar branches use 12 AWG for both (+) and (–). Standardized from original mixed 16/18 AWG for marine durability and inventory simplification.

ID	Branch	To	Gauge	Length	Fuse	Notes
P7/P8	Starlink (direct)	Starlink Mini DC	12 AWG	18–24"	ATO 7.5A SB	No converter. 12–48V. Long: plate→top deck.
P9/P10	5V Pwr Module	Pwr Mod input	12 AWG	4–6"	ATO 2A SB	Short on plate. Feeds Pixhawk.
P11/P12	12V Buck	Buck input	12 AWG	4–6"	ATO 3A SB	Short on plate. Feeds Jetson+fan.
P13/P14	5V Buck	Buck input	12 AWG	4–6"	ATO 1A SB	Short on plate. Feeds ESP32.
P15/P16	BEC (Castle)	BEC input	12 AWG	4–6"	ATO 2A SB	Short on plate. Feeds servo.

All (–) branches connect to ATO block negative bus = electrically same as (–) busbar. Single ground reference, no separate ground wiring needed.

2.4 — Load-Side Connections (Converter → Device)

ID	From	To	Gauge	Length	Connector
P18	5V Power Module	Pixhawk 4X FC	22 AWG	4"	DF13 6-pin
P19	12V Buck	Jetson Orin Nano	22 AWG	6"	DC barrel
P20	12V Buck	Cooling Fan	22 AWG	4"	JST 2-pin
P21	5V Buck	ESP32-S3 (breakout)	22 AWG	4"	Screw terminal
P22	BEC (Castle)	Rudder Servo	22 AWG	18–24"	Servo 3-pin
P23	ESP32 3.3V	Sensors (all)	26 AWG	varies	DuPont/JST
P24	Jetson 3.3V	Camera IMX219	Ribbon	3"	15-pin FPC

2.5 — Fuse Coordination (Max Load < Fuse < Wire Ampacity)

Branch	Max Load	Fuse	Wire	Ampacity	Status
Trunk (prod)	61.3A	100A	4 AWG	135A	PASS
Trunk (demo)	40A gov	40A	10–12 AWG	41–55A	PASS
ESC/Motor	54A	60A	8 AWG	73A	PASS
Starlink	4.7A	7.5A	12 AWG	41A	PASS
Pixhawk	0.5A	2A	12 AWG	41A	PASS
Jetson+Fan	1.26A	3A	12 AWG	41A	PASS
ESP32	0.23A	1A	12 AWG	41A	PASS
BEC/Servo	1.0A	2A	12 AWG	41A	PASS

3 — Signal & Data Wiring

3.1 — Compute-Zone Interconnects (Short Runs, <6")

ID	From	To	Gauge	Protocol	Length	Connector
S1	ESP32 UART1	Pixhawk (IF-1)	22 AWG	MAVLink UART 3.3V	4–6"	JST-SH 4-pin
S2	ESP32 UART2	Jetson (IF-3)	22 AWG	UART 115200 3.3V	4–6"	JST-SH 4-pin
S3	Pixhawk PWM	ESC Signal	22 AWG	PWM 50Hz	4–6"	Servo 3-pin
S4	Jetson CSI	Camera IMX219	Ribbon	MIPI CSI-2 LVDS	3"	15-pin FPC
S5	Jetson Eth	Starlink Mini	Cat5e/6	100BASE-T	12–18"	RJ-45
S7	ESP32 I2C	MPU6050 IMU	26 AWG	I2C 3.3V	3–4"	DuPont 4-pin

S3 note: PWM to ESC should come from **Pixhawk**, not ESP32 direct GPIO. Pixhawk is confirmed for ArduPilot Rover mode. Confirm pin assignment with Robert.

3.2 — Bow Sensor Bundle via Shielded CAT6 Cable

Shielded CAT6 patch cable runs from ESP32 area through hull to external bow sensor mount (submerged). Carries **4 signal lines** using all 4 twisted pairs along the same path. The 18–24" run through the electrically noisy hull benefits from the CAT6's foil shield plus per-pair twisting — far superior to copper-tape-wrapped individual 26 AWG, especially for the analog TDS signal and I2C clock/data.

Pair	Signal Line (ID)	Wire A	Wire B	Destination Sensor
1	S8 — pH I2C SDA	I2C SDA	GND	Atlas EZO-pH + isolation board
2	S8 — pH I2C SCL	I2C SCL	3.3V VCC	Atlas EZO-pH + isolation board
3	S9 — Temperature	1-Wire DATA	GND	DS18B20 (waterproof)
4	S10 — TDS Conductivity	Analog signal	GND	TDS module (0–3.3V)

CAT6 shield grounding: Ground the foil shield at the **ESP32 end only** (source end). Leave bow end unconnected. Solder/crimp drain wire to ESP32 GND at breakout board.

NOT in the CAT6: S7 (IMU — 3–4" on plate, short run), S11 (GPS — top deck, different path), S12 (Ultrasonic — digital I/O, lower sensitivity).

3.3 — Remaining Sensor Lines (Individual Runs)

ID	From	To	Gauge	Protocol	Length	Connector
S11	ESP32 UART	GPS NEO6MV2	26 AWG	UART 9600 3.3V	12–18"	DuPont 4-pin
S12	ESP32 GPIO	DFRobot Ultrasonic	26 AWG	Digital I/O	18–24"	DuPont 4-pin

S11 routes to top deck. Keep ≥10cm from GPS antenna to Starlink/ESP32 antenna. S12 routes to bow along hull wall, separated from motor power lines.

4 — EMI Filtering Placement

4.1 — Busbar Bypass Caps

Part	Value	Placement	Status
C1	1000uF / 35V+ electrolytic	Across (+)/(–) busbar studs, leads <1"	ORDER
C2	100nF / 50V+ ceramic	Parallel with C1, same studs	ON HAND (verify 50V+)

Solder C1+C2 in parallel, ring terminals to studs. Zip-tie for vibration.

4.2 — Per-Branch Ferrites & Caps

Branch	Input Ferrite	Output Caps	Status
ESC/Motor (P5)	F1: Hondark toroid (38×19×13mm) on 8 AWG (+), <2" from ESC	C3: 470uF LOW-ESR + C2 at ESC input	ORDER F1, C3
5V Buck→ESP32 (P13/P21)	F2: 13mm clip-on on 12 AWG (+)	F4: clip-on on 22 AWG pair; C4: 10uF + C2 at ESP32 VIN	ORDER F2-F4, C4
12V Buck→Jetson (P11/P19)	F2: 13mm clip-on on 12 AWG (+)	C5: 100uF/50V + C2 at Jetson barrel	C5 ON HAND
Starlink (P7)	F3: 13mm clip-on on 12 AWG (+)	C7: 100uF/50V + C2 at DC plug	C7 ON HAND
BEC→Servo (P15/P22)	— (none)	C6: 100uF/50V + C2 at servo	C6 ON HAND
Pixhawk (P9)	— (none)	— (none)	Internal filtering

Rules: Clip-ons on (+) wire only, except F4 (both wires, common-mode). F1 toroid = single pass only. Low-ESR critical at ESC position only.

4.3 — Sensor Signal Filtering

Signal	Filter	Placement	Priority
S10 (TDS)	100nF ceramic pin-to-GND	<0.25" from ESP32 ADC pin	CRITICAL
S7 (IMU I2C)	BOJACK bead on SDA + SCL	<1" from ESP32 I2C pins	HIGH (2 beads)
S8 (pH I2C)	BOJACK bead on SDA + SCL	ESP32 end of CAT6 breakout	HIGH (2 beads)
S9 (1-Wire)	100nF ceramic pin-to-GND	<0.5" from ESP32 pin	Optional
S11 (GPS)	100nF ceramic pin-to-GND	<0.5" from ESP32 pin	Optional

Total ferrite beads for I2C: 4 (2 for S7 SDA/SCL + 2 for S8 SDA/SCL).

4.4 — Copper Tape Shielding

Cable	Priority	Shield Drain	Notes
P21: 5V Buck→ESP32	HIGH	5V buck GND (source)	Wrap before loom
CAT6 bundle (S8/S9/S10)	—	Foil shield handles it	No extra tape
S11: GPS UART	MEDIUM	ESP32 GND	Longer top-deck run
S1/S2: UART lines	MEDIUM	ESP32 GND	Short, optional
P5/P6: Motor 3ϕ	SKIP	—	Noise source
P1–P4: Trunk	SKIP	—	Short, low-Z

Rules: (1) Ground ONE end only (source). Ungrounded = antenna. Both ends = ground loop. (2) Layer: copper tape → auto tape → split loom. (3) Leave 0.5" bare at connectors.

5 — Wire Routing Channels

Channel	Wires	Path
Centerline (hull floor)	4 AWG trunk, 8 AWG motor	Battery→busbars; ESC→motor. Dedicated loom.
Port hull wall	12 AWG branches (P7–P16)	Busbars→converters. Short runs.
Starboard hull wall	22–26 AWG signals, CAT6	ESP32→bow/top deck. Separated from power.
Vertical / top deck	P7/P8 Starlink, S5 Eth, S11 GPS	Plate→top deck pass-throughs

Critical separation: Signal wires (22–26 AWG, CAT6) must stay ≥12" from 8 AWG motor power. Route on opposite hull walls.

6 — Assembly Sequence

1. Verify all connections de-energized.
2. Build trunk: P1–P3 (MRBF→kill switch→(+) busbar), then P4 ((–) busbar).
3. Busbar bypass caps: C1+C2 in parallel on studs. Leads <1". Zip-tie.
4. Install ATO fuses in block (leave MRBF and MIDI empty).
5. Wire all 12 AWG branches P7–P16 from busbars to converters.
6. Input ferrites: F1 toroid on P5, F2 clip-ons on P11 and P13, F3 clip-on on P7.
7. Wire motor branch P5/P6 through MIDI block to ESC.
8. Wire load-side P17–P24 (converters→devices).
9. Output ferrite F4 on P21 (22 AWG pair to ESP32).
10. Output caps: C3+C2 at ESC, C4+C2 at ESP32, C5+C2 at Jetson, C6+C2 at servo, C7+C2 at Starlink.
11. CAT6 bow sensor bundle: route from ESP32 to bow. Break out pairs. Ground shield at ESP32 GND only.
12. Sensor filtering: C2 on TDS ADC pin (S10). F5 beads on I2C lines (4 total).
13. Remaining signals: S1, S2, S3, S7, S11, S12. Copper tape per priority table.
14. Final inspection: all terminals tight, polarity correct, no stray strands, no cap shorts.
15. Insert MIDI 60A fuse element.
16. Insert MRBF fuse element LAST (this energizes the system).
17. Multimeter test at each load point before powering devices.

7 — Wire Cut List

Cut and label all wires before starting assembly.

Gauge	Color	Cut Length	Qty	Labels / Purpose
4 AWG	RED	8"	3	P1, P2, P3 (trim to 6" after crimp)
4 AWG	BLK	14"	1	P4 (trim to 12")
8 AWG	RED	8"	1	P5 (busbar→ESC +)
8 AWG	BLK	8"	1	P6 (busbar→ESC –)
8 AWG	any ×3	20"	3	P17 three-phase ESC→motor
12 AWG	RED	8"	4	P9, P11, P13, P15 (short branch +)
12 AWG	BLK	8"	4	P10, P12, P14, P16 (short branch –)
12 AWG	RED	26"	1	P7 Starlink + (long, trim to 18–24")
12 AWG	BLK	26"	1	P8 Starlink – (long)
22 AWG	pair	6"	4	P18, P19, P20, P21 (load-side power)
22 AWG	pair	24"	1	P22 BEC→servo (long run to stern)
22 AWG	pair	6"	3	S1, S2, S3 (compute-zone signals)
26 AWG	4-wire	5"	1	S7 IMU (SDA, SCL, VCC, GND)
26 AWG	2-wire	20"	1	S11 GPS (TX, RX + VCC/GND)
26 AWG	4-wire	24"	1	S12 Ultrasonic (separate bow run)
CAT6	—	30"	1	Bow sensor bundle (S8/S9/S10)
10–12 AWG	RED+BLK	8" ea	1 ea	D1/D2 demo adapter pigtail

8 — Pending Items

- ■ Confirm S3 PWM source with Robert (Pixhawk, not ESP32 direct GPIO)
- ■ Verify Starlink Mini DC connector type — may need custom pigtail
- ■ Confirm pH isolation board ordered (Atlas EZO-pH galvanic protection)
- ■ Install EMI parts when arrived; no extended motor testing until F1 (ESC toroid) in place
- ■ Motor inrush test — confirm MIDI 60A holds during pool startup (70A if nuisance trips)
- ■ Jetson boot inrush test — confirm ATO 3A holds
- ■ Waterproofing / conformal coating plan TBD